A Comparative Evaluation on Melody Generation of Large Language Models

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***Abstract*—Music has always been central to human culture. With the rise of digital tech and smart devices, its role in the Consumer Electronics (CE) sector has grown. Recently, large-scale language models (LLMs) like ChatGPT and Bard are promising to change the way we compose music. This paper explores the composing ability of Large Language Models (LLM) such as ChatGPT and Bard. Unlike older methods that required deep musical and statistical knowledge, LLM lets users simply describe what they want in music. We evaluated this LLM on how well it matches user input and the overall quality of its compositions, looking at factors like repetitiveness and scale diversity. This research aims to highlight the strengths, uses, and limits of LLM in music, setting the stage for its greater role in CE and the wider music world.**

Generated Melodies: https://github.com/545659928/LLMMelody

***Index Terms*—Melody Generation, Large Language Model, Music Creation, Text to Music**

I. Introduction

Music has been central to human culture for ages. The entertainment value makes it being crucial in the Consumer Electronic (CE) sector [1], especially with the growth of digital technology. The CE sector has not only seen a rise in demand for quality sound systems but also in software and applications that play or generate music. Music apps, streaming services, and even machine composition tools [2] are now integral parts of the CE landscape. As a starting point for music composition, while traditional music creation requires profound knowledge and talent, the evolving CE landscape reveals that machine composition technology is opening the door to this art is democratizing access to this art.

Machine composition involves creating music with computers, with early methods like Iannis Xenakis [3]'s based on music parameters and probability distribution. This approach required deep statistical and musical knowledge. However, modern machine composition tech has reduced the expertise needed. AI-driven tools fall into rule-based systems, such as Google's Magenta AI [4], which uses musical parameters including pitch, rhythm, and chord progressions to generate music; and text-to-music (T2M) features like MusicLM [5], which transforms text into musical elements via the T2M tool. Despite AI simplifying the process, a basic musical understanding is still crucial for users.

Building on this evolution, Large Language Models (LLMs) like OpenAI's GPT [6] series and Google's Bard are now at the forefront of technological attention. These models have demonstrated huge capabilities beyond text prediction, extending their influence into fields like music composition. Users can state their requirements using plain language, enabling the LLMs to produce corresponding musical pieces. LLM-based composition services are growing in popularity, potentially transforming the music industry by attracting a wider audience to music creation. This is promising to refresh global musical culture and drive the CE sector's evolution.

However, to the best of our knowledge, while endeavors and researches using LLMs for music composition [7], [8] are undoubtedly on the rise, a comprehensive evaluation of its composing capability remains unexplored. The lack of evaluation presents several potential issues. Firstly, the ambiguity of its utility means we cannot accurately discern the true value, application range, and limitations of the technology. This ambiguity also complicates determining the actual scope of LLMs technology application and its market positioning. Furthermore, if the utility of the technology is not clear, it becomes challenging to establish policies for service expansion or product development. Decisions regarding the design of specific services, targeting user groups, and even pricing become problematic. The step of evaluating the technology is indispensable for translating research outcomes into real products or services.

Facing the challenges posed by the ambiguity of utility in using LLMs for music, we sought to provide a definitive evaluation of their composing abilities by inputting prompts with both specific musical features and abstract features. Due to the complexity of the arrangements and voice synthesis for lyrics, this paper only focuses on the LLM's ability to compose fundamental melodies. Specifically, LLMs are used

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to generate melodies in the ABC-notation format, a text- and symbol-based format for music notation that uses letters for pitches and numbers for duration.

In assessing the outputs, our approach is twofold. From an objective standpoint, we look into: 1) how closely the melodies adhere to the provided input prompts, 2) their capability to offer a balanced listening experience, and 3) their representation of emotional and thematic content. Subjectively, we rely on participants' feedback, gauging their sense of listening balance and emotional and thematic connection.

II. Background and Related Work

A. Large Language Models

The research landscape in AI is undergoing a revolutionary transformation due to the rapid advancements of Large Language Models (LLMs). Models like OpenAI's GPT series and Bard have proven their capability across a multitude of linguistic tasks, including but not limited to question-answering systems. These LLMs are emerging as universal solvers for language-based tasks, and research efforts are now centered on utilizing their potential. The effectiveness of these models rests on two cornerstones: a large dataset for training and sophisticated model architectures. After being trained billions of parameters and copious amounts of textual data [9], LLMs can generate and understand consistent, precise, and fluid text, a capability that was out of reach using conventional NLP techniques.

B. Text to Music

AI-based music composition tools can be broadly categorized into two types: (i) Rule-based systems: These systems allow for composition based on musical concepts like tonic, beat, etc. As a result, they demand substantial musical knowledge. Examples include Magenta AI developed by Google. (ii) Tools with Text to Music feature: These systems allow users to define musical details using text, enabling even those without musical knowledge to compose based on nuance. MusicLM [5] is an example. In MusicLM, user-inputted text descriptions are converted into musical features using the Transformer for Text-to-Music (T2M) Transfer tool. MusicLM extracts five musical features from the text: melody, harmony, rhythm, song structure, and acoustical properties. While this method offers a comprehensive approach, it requires users to have knowledge of the music features extracted by MusicLM.

III. Evaluation

This study aims to understand how well LLMs can generate music based on following three research questions(RQ).

RQ1: *Are the quantifiable specifications provided in the LLMs input, such as song length and beat equivalency, accurately reflected in the generated melody?*
This study specifies 8 measures long and a 4/4 beat. Ensuring that LLMs generate melodies with the musical elements designated in the input is paramount.

RQ2: *Can LLMs accurately recognize and generate melodies in alignment with the user's specified abstract prompts?*
Recognizing and interpreting abstract prompts to produce coherent melodies is a critical assessment of LLM's capabilities. The challenge is not just about generating any music but ensuring that it is equivalent with the intended emotional and thematic conveyed by the user's instructions.

RQ3: *Can LLMs provide a balanced listening experience that meets users' expectations when generating music?*
This study aims to assess the LLMs model's ability to generate melodies from specified prompts and ensure they offer a balanced listening experience for users. By evaluating the LLM's performance, we seek insights into its potential and challenges in automated music creation.

A. Experiment Setting

1) Model Choice and Prompt Design: In this study, we evaluate two LLMs for melody generation: OpenAI's ChatGPT (GPT-4.0 August 3 Version) and Google's Bard (v1.0.0 July 13 Version). These models have state-of-the-art performance and prominence in the domain of language-based machine learning. When instructing these models to generate melodies, we employ a structured prompt format, utilizing the notation — "pitch" duration — for individual notes. This prompt framework not only dictates specific rules for the melody but also designates the genre as rock, defines the atmosphere as either upbeat(happy) or melancholic(sad), and sets the piece's length at 32 counts, measured in 4-beat equivalents. Detailed examples and further specifics can be found in Figure 1.We generated five melodies each for both [happy] and [sad] atmosphere using the LLMs to produce diverse outputs.

2) Data Processing: To gain a deeper understanding of LLM's compositional capabilities, we employed a series of data processing techniques, as illustrated in Fig. 2. First, LLMs generates melody in the ABC-notation format based on the given prompt. Second, we evaluate them from objective/subjective dimensions. Objectively, we analyzed various musical features of the compositions, uncovering inherent attributes such as the number of repetitions. Subjectively, using the music21 framework, we converted the ABC-notation into MIDI files, allowing listeners to play them directly.

3) Objective Metrics: Drawing inspiration from the works of [10], [11], we conducted a multi-faceted evaluation of various MIDI sequences. For repetitiveness, ChatGPT exhibited higher 2-MIDI and 3-MIDI repetitions compared to Bard. They control user listening experience as too less repetitive leads to tire and too much lead to monotony. Average chord tonal distance metric in a brief melody significantly shapes the song's atmosphere and progression. Unique MIDI numbers indicates the total unique MIDI numbers within a sequence. In brief melodies, the total unique MIDI numbers in a sequence can help understand a song's genre and stylistic depth. Beat consistency confirms whether beat of the generated melody matches to the phrase specified by prompt (4/4). Lastly, phrase consistency confirms whether phrase of the generated melody matches to the phrase specified via prompt (8 phrases).

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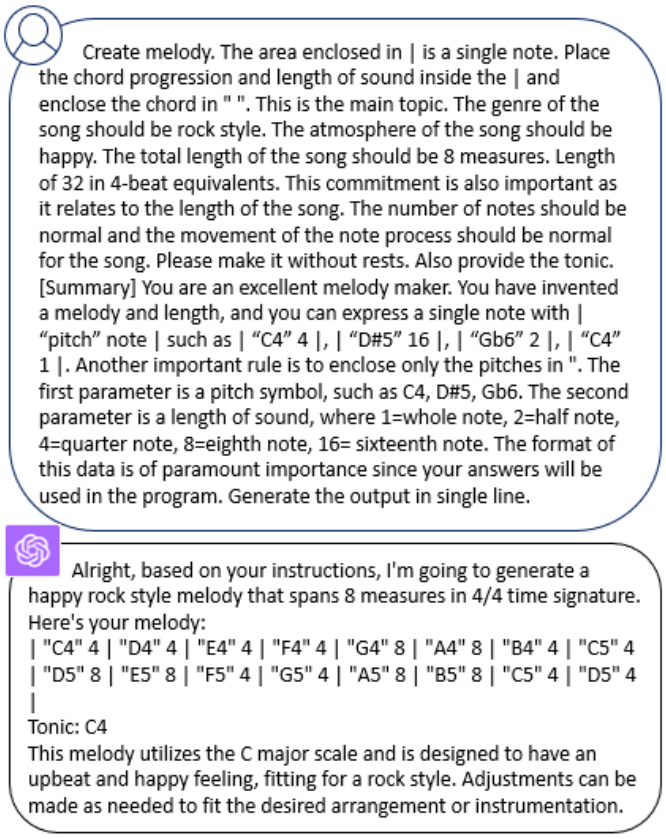

Fig. 1. Example of ChatGPT's prompt and its reply

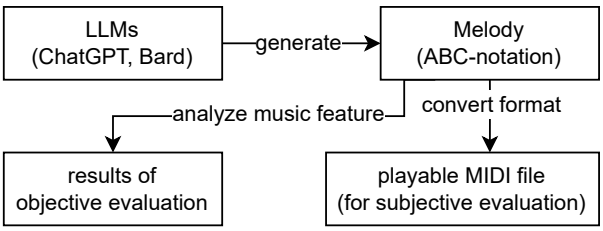

Fig. 2. Procedure of data process

These metrics would illustrate the nuanced capabilities and challenges facing each LLM in music generation.

4) Subjective Metrics: As shown in Table I, within the subjective metrics section, we primarily evaluate the model's output based on two criteria: atmosphere consistency acknowledgment and genre consistency acknowledgment. To verify if LLMs truly understand the nuances of musical atmosphere and genre, we have established a cross-experimental setup among the LLMs. Music generated by one model is assessed both by itself and by another, aiming to confirm LLMs' recognition of the specified atmosphere and genre. Atmosphere consistency is rated by LLMs from 1 (most sad) to 10 (most happy), while genre consistency ranges from 1 (least rock) to 10 (most rock). Subjective evaluation by humans was conducted with 19 participants (average age: 30.5; male: 63.2%). After their auditory experience, participants were asked to provide feedback on the melodies. They were prompted to rate atmosphere consistency (ranking melodies on a scale of 1-5, with 5 being strongly happy/sad), genre consistency (ranking melodies on a scale of 1-5, with 5 indicating a strong resemblance to rock), and preference (indicating which LLM they favored). Participants were also asked for the reasons behind their preferences.

B. Results and Discussions

RQ1 was examined by objective evaluation to assess how accurate LLMs generate melody with specified musical features, and how natural generated melodies perceived. For RQ1, our analysis points towards ChatGPT's superior natural language processing capabilities in interpreting and executing on provided musical specifications, beat consistency and number of phrases. Although both LLMs have 100% beat consistency, data for number of phrases for ChatGPT (7.6) is higher than Bard. (6.2). So ChatGPT is more consistent in matching the musical details with the input prompts.

RQ2 to examine abstract prompts recognition also priors ChatGPT with higher natural language processing ability. Difference in Unique MIDI numbers indicates the different atmosphere, which advocates that LLMs trying to implement processed atmosphere. Difference is higher for ChatGPT (0.6) than Bard (0.2). In our subjective evaluation, melodies generated by ChatGPT were not only assessed by ChatGPT itself but also by Bard, and vice versa, providing a cross-examination. Addressing RQ2, ChatGPT consistently demonstrated superiority in generating atmospherically accurate melodies. For happy melodies, the scores were: ChatGPT's own evaluation (7.0/10.0), Bard's evaluation of ChatGPT (5.6/10.0), and participants for ChatGPT (3.8/5.0). Conversely, Bard's happy melodies were evaluated as: by ChatGPT (6.0/10.0), by Bard (5.8/10.0), and participants (2.8/5.0). Generally, ChatGPT generated happier melody, and all scores show the happy atmosphere. Cross-evaluation presents each LLMs evaluate own melody higher than opponents. However, a shift was noted for the sad atmospheres. The scores for melodies produced by ChatGPT were: ChatGPT (4.3/10.0), Bard (5.2/10.0), and participants (3.5/5.0). In contrast, Bard's sad melodies received scores of: ChatGPT (6.4/10.0), Bard (4.8/10.0), and participants (2.9/5.0). Cross-evaluation in sad atmospheres also presents own melody higher than opponents. For sad scores, except small LLMs results admit sad melody, most LLMs and all participants defines melody to be happy. It illustrates that LLMs struggled to generate sad atmosphere melodies.

The melody genre recognition was examined in the same method. For melodies produced by ChatGPT, the scores were: evaluated by ChatGPT (6.9/10.0), evaluated by Bard (3.9/10.0), and by participants (2.9/5.0). In comparison, for Bard-produced melodies: evaluated by ChatGPT (7.3/10.0), by Bard itself (5.3/10.0), and by participants (4.3/5.0). These scores reveal an interesting trend: while ChatGPT gave a higher score to Bard's rendition of the rock genre than its own, Bard's evaluation of ChatGPT's work was significantly lower. Moreover, the human participants favored Bard's rendition over ChatGPT's. This cross-evaluation indicates that Bard may possess a more intuitive understanding or better training data

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TABLE I
COMPARISON OF OBJECTIVE AND SUBJECTIVE METRICS FOR MELODY GENERATION ACROSS LLMS

	ChatGPT-Happy	ChatGPT-Sad	ChatGPT	Bard-Happy	Bard-Sad	Bard
Objective Metrics						
2-MIDI numbers repetitions	11.2	12.6	11.9	7.4	9.2	8.3
3-MIDI numbers repetitions	6.8	8.8	7.8	2.2	4.6	3.4
Average Chord Tonal Distance	2.08	1.92	2.00	2.39	2.15	2.27
Unique MIDI numbers	7.4	8.2	7.8	7.6	7.8	7.7
Beat Consistency	100%	100%	100%	100%	100%	100%
Number of Phrases (8 on prompt)	6.8	8.4	7.6	6.0	6.4	6.2
Subjective Metrics						
atmosphere consistency acknowledge -ChatGPT (Out of 10)	7.0	4.3	——	6.0	6.4	——
Genre consistency acknowledge -ChatGPT (Out of 10)	——	——	6.9	——	——	7.3
atmosphere consistency acknowledge -Bard (Out of 10)	5.6	5.2	——	5.8	4.8	——
Genre consistency acknowledge -Bard (Out of 10)	——	——	3.9	——	——	5.3
atmosphere consistency acknowledge -Human (Out of 5)	3.8	3.5	——	2.8	2.9	——
Genre consistency acknowledge -Human (Out of 5)	——	——	2.9	——	——	4.3
Preference -Human	——	——	26%	——	——	74%

for the rock genre. Overall, these results support the conclusion that Bard more effectively captures the nuances of the rock genre, consistent with our objective evaluation.

However, some participants expressed that there was "honestly nothing rock about pieces played by piano," hinting at the limitations of generating only melodies without complementary instrumentation. This suggests a more adept capability on ChatGPT's part in generating atmospheres that align with the specified prompts.

For RQ3, Bard displayed superior melody expressiveness evidenced from two significant metrics. A critical aspect to consider is the number of repetitions; while a high count can make melodies feel monotonous, too low might tire the listener. In this study, both LLMs produced high repetition counts for the length of their melodies, suggesting a risk of monotony. Bard's counts—8.3 and 3.4 for 2-MIDI and 3-MIDI—are more optimal compared to ChatGPT's 11.9 and 7.8, implying a potentially better listening experience with Bard. Additionally, a higher average chord tonal distance, like Bard's 2.27 versus ChatGPT's 2.00, indicates a more dynamic chord progression, fitting well for genres like rock. Together, these metrics suggest Bard offers a more balanced listening experience, melding optimal repetition with dynamic tonal shifts. From participants, 74% favored Bard's compositions. This subjective viewpoints also confirms objective measures.

IV. CONCLUSION AND FUTURE WORK

In this paper, our evaluations reveal differences in the compositional abilities of ChatGPT and Bard: ChatGPT is better at following specific musical features, while Bard is stronger in emotional and genre-specific expression. Still, ChatGPT did well in capturing the intended atmosphere from abstract prompts, and Bard was good at representing the rock genre. User feedback showed a preference for emotional depth and musical richness over strict adherence to the initial prompts. This highlights the balance between computational accuracy and musical creativity as a focus for future research in machine-aided music composition.

In our future research, we plan to explore three main areas: (1) Prompt Optimization – We'll investigate how different prompts affect the music produced by the LLM and refine them for better results. (2) Holistic Evaluation – Instead of just focusing on melody, we aim to include accompaniments, allowing for a broader comparison with traditional music generation methods. (3) User-Centric Tools – Recognizing the potential of LLM in music, we plan to create tools that help beginners compose intricate pieces with AI support.

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